



«Approved»
by the Dean
Syzdykova Z.A.
« » of 2024

Syllabus Academic Year 2024-2025

1. General Information	
Course title	Calculus
Degree cycle (level)/major	Bachelor, Group of specialties ITManagement, Digital Journalism, IT Entrepreneurship
Year, term	1,1
Number of credits	5
Language of delivery:	English
Prerequisites	Secondary school math courses
Postrequisites	Theory of Probability and Statistics
Lecturer(s)	Rakhymzhan Kazbek, PhD candidate in Applied Mathematics, Assistant Professor, Astana IT University, Expo, C1 block, 3rd floor, office C1.1.336 r.kazbek@astanait.edu.kz .
2. Goals, Objectives and Learning Outcomes of the Course	
1. Course Description	Calculus is a foundational course at Astana IT University; it plays an important role in the understanding of science, engineering, economics, and computer science, among other disciplines. This Calculus course covers differentiation and integration of functions of one and multiple variables, with applications. Topics include: • Concepts of Function, Limits and Continuity • Differentiation Rules, Applications, Rates, Approximations, and Extremum Problems • Definite and Indefinite Integration • The Fundamental Theorem of Calculus • Applications to Geometry: Area, Volume, and Arc Length • Techniques of Integration • Approximation of Definite Integrals, Improper Integrals, and L'Hôpital's Rule • Multivariable functions, scalar field • Gradient and its properties • Applications Double and concept of multidimensional integrals
2. Course Goal(s)	After completing this course, students should have developed a clear understanding of the fundamental concepts of single and multivariable variable calculus and a range of skills allowing them to work effectively with the concepts. The main goal is to be able to apply above-mentioned

	tools to problems in prerequisites courses.
3. Course Objectives:	After completing this course, students should demonstrate competency in the following skills: • Use both the limit definition and rules of differentiation to differentiate functions. • Sketch the graph of a function using asymptotes, critical points, the derivative test for increasing/decreasing functions, and concavity. • Apply differentiation to solve applied max/min problems. • Apply differentiation to solve related rates problems. • Evaluate integrals by using the Fundamental Theorem of Calculus. • Evaluate integrals using substitution and integration by parts. • Use L'Hospital's rule to evaluate certain indefinite forms. • Determine convergence/divergence of improper integrals and evaluate convergent improper integrals. • Compute the gradient of the multidimensional function. • Evaluate elementary double integrals and apply them to compute areas and volumes.
4. Skills & Competences	•
5. Course Learning Outcomes:	The successful Calculus student should understand and be able to apply concepts related to limits, continuity, derivatives and basic integrals. The student should be able to work with these concepts numerically, graphically and analytically.
6. Methods of Assessment	• Homework • Midterm, including Classwork, Pre-Final and Final exams...
7. Reading List	Assigned reading materials and presentations should be read prior to class. Class lectures and discussions will proceed with supplemental and advanced topics, which could be difficult to understand unless students have read the assigned material. Readings are listed in the schedule section. All necessary updates and / or changes to the course will be reflected in the Learning Management System (moodle.astanait.edu.kz). <u>Basic Literature:</u> 1. Thomas' Calculus. By George b. Thomas, revised by J.Hass, C.Heil, M.D.Weir, Pearson Publishing Company. 14n edition 2. George b. Thomas, Jr., Ross L. Finney, Calculus and Analytic Geometry. Part II. Addison-Wesley Publishing Company. 9 th edition. <u>Supplementary literature:</u> 1. G. N. Berman, A collection of problems on a course of Mathematical Analysis 2. Г.М.Фихтенгольц. Основы математического анализа, Т.1, Изд-е 9-ое, Изд. Лань – 2008. – 448 с.
1. Resources	1. https://ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-042j-mathematics-for-computer-science-fall-2010/

2. Course policy	Course and University policies include: Attendance: Attendance is not allocated any grading points in the marking scheme, but is compulsory to pass the course. Normally students are required to achieve course attendance of minimum 70% to get admitted to the examination rubric. In case a student misses 30% or more class sessions without a valid excuse the instructor has the right to mark him as “not graded”. In such case a student is not admitted to the exam and automatically fails the course. In cases, when a student misses class session due to valid reasons (is excused by the instructor or the dean’s office) he or she has to confirm the absence reason using a valid document in accordance with the academic policy of AITU. It should be NOTED that in cases when a student is excused for 30% of the scheduled class sessions or more he or she has to study material provided under the course on their own. Course instructor might provide additional opportunities to submit missed graded pieces of work during office hours or conduct alternative assessment exercises using method of his or her choosing. Preparation for Class: Class participation is a very important part of the learning process in this course. Although not explicitly grade, students will be evaluated on the QUALITY of their contributions and insights. Quality comments possess one or more of the following properties: - Offers a different and unique, but relevant, perspective; - Contributes to moving the discussion and analysis forward; - Builds on other comments. Class work: The duration of each lecture and practical lesson is 50 minutes for offline class, and 40 minutes for online class. Students are expected to complete all readings and assignments ahead of time, attend class regularly and participate in class discussions. In case of systemic student’s misconduct, the student can be dispensed from the classes. Being late on class: When students come to class late, it can disrupt the flow of a lecture or discussion, distract other students, impede learning, and generally erode class morale. Moreover, if left unchecked, lateness can become chronic and spread throughout the class. Therefore, the being late to the class is not welcome and can have restriction activities by the course instructor. Attestation I and II: Students, who score less than 25% for Attestation period I or Attestation period II (RK1/RK2) automatically fail the course.
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Home work / Assignments: The assignments are designed to acquaint students with the theoretical knowledge and practical skills required for the course. The textbook readings will be supplemented with materials collected from recent professional articles and journals. In case of using someone's work (papers, articles, any publications), all works must be properly cited. Failure to cite work will be resulted as a cheating from the students and may be a subject of additional disciplinary measures.

Late submissions: Most assignments will be discussed in class on the due date. It is expected that all work will be submitted on time. All gradings are based using a percentage grading scale.

In the case of some extraordinary event, students should notify the course instructor and request an extension of the deadline for submission. If approved, a new date will be given to the student depending upon the circumstances by the instructor.

Final exam*

Laptops and mobile devices can only be used for classroom purposes when directed by the course instructor. Misuse of laptops or handheld devices will be considered a breach of discipline and appropriate action will be initiated by the instructor.

Online lessons can be used in case if there won't be a chance to make offline traditional lessons. It must not discourage the interest and enthusiasm of students. The main software to run the online lessons is Microsoft Teams for video calls and live webinars, and Moodle (moodle.astanait.edu.kz) as a Learning Management System. Also, some alternatives such as Telegram, Zoom, or other messenger may be involved as an additional workaround.

Cheating and plagiarism are defined in the Academic conduct policies of the university and include:

1. Submitting work that is not your own papers, assignments, or exams;
2. Copying ideas, words, or graphics from a published or unpublished source without appropriate citation;
3. Submitting or using falsified data;
4. Submitting the same work for credit in two courses without prior consent of both instructors.

Any student who is found cheating or plagiarizing on any work for this course will receive 0 (zero) for that work and further actions will also be taken regarding academic conduct policies of the university.

Academic Conduct Policies of the university: The full texts of all the academic conduct code will be posted to the students using Learning Management System (moodle.astanait.edu.kz).

Contacting the Course instructor: The easiest and most reliable way to get in touch with the course instructor is by email. Students must feel free to send email if you have a question related to the course. Instructor responds as soon as they can but not always instantaneously. Besides

	that, students are also welcomed to arrange a one-to-one meeting with the instructor by their office during office hours to discuss the class using both offline and online.
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** Final exam can be given by instructor in the any other forms such as Written Exam, Final Quiz, Oral Tasks and Exams, etc.*

3. Course Content

#	Abbreviation	Meaning
1	ISIS	Instructor-supervised independent work
2	SIS	Students' independent work
3	IP	Individual project
4	PA	Practical assignment
5	LW	Laboratory work
6	MCQ	Multiple choice quiz

3. Course Content

3.1 Lecture, Practical/Seminar/Laboratory Session Plans

Week No	Course Topic	Lectures (H/W)	Practical sessions (H/W)	Lab. sessions (H/W)	TSIS (H/W)	SIS (H/W)
1	Functions and their graphs. 1. Definition. Domain and Range. Elementary functions, Cartesian coordinates. Graphs. Shifts, and Scaling graphs. 2. Elementary functions. Trigonometric and Transcendental Functions. 3. Inverse Functions. The composition of functions	3	2			
2	Limits and Continuity. 1. Sequences. Definition of a limit. 2. Sequences converging to a limit and their properties 3. The precise definition of a Limit	3	2			
3	Limits and Continuity. 1. One-sided limits 2. Continuity 3. Limits involving infinity. Relative Rates of growth.	3	2			

4	Derivatives. 1. Tangent Lines and the Derivative at a Point 2. The Derivative as a Function 3. Differentiation Rules	3	2			
5	Derivatives. 1. Derivatives of Trigonometric Functions 2. The Chain Rule 3. Linearization and Differentials	3	2			
	Midterm					
6	Applications of Derivatives. 1. Extreme Values of Functions on Closed Intervals 2. The Mean Value Theorem 3. Concavity and Curve Sketching 4.	3	2			
7	Definite integrals. 1. Area and estimating with finite sums. 2. The fundamental theorem of Calculus. 7.3 Indefinite Integrals and the Substitution Method 3. Definite Integral Substitutions and the Area Between Curves.	3	2			
8	Techniques of Integration 1. Using Basic Integration Formulas 2. Integration by part. 3. Trigonometric Integrals 4.	3	2			
9	Techniques of Integration , Series 1. Trigonometric Substitutions 2. Integration of Rational Functions by Partial Fractions 3. Sequences 4. Infinite Series 5. The Integral Test 6. Comparison Tests	3	2			
10	Infinite Sequences and Series 1: Absolute Convergence; The Ratio and Root Tests 2. Power Series 3. Taylor and Maclaurin Series 4.	3	2			

	Total hours: 50	30	20			
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3.2 List of Assignments for Student Independent Study

Nº	Assignments (topics) for Independent study	Hours	Recommended literature and other sources (links)	Form of submission
1	2	3	4	5
1	Graphing with Software. Learn how to use MATLAB to solve problems. Additional and Advanced Exercises in Textbook.	30	[2], Chapters 1-8	Exercises

4. Student Performance Evaluation System for the Course

Period	Assignments**	Number of points	Total
1 st attestation	a. 2 Homeworks	20	30
	b. Quiz	20	
	c. Midterm	60	
2 nd attestation	a. 2 Homeworks	20	30
	b. Quiz	20	
	c. Endterm	60	
Final exam *	A written exam	100	40
Total	0,3*1stAtt+0,3*2ndAtt+0,4*Final		100

****** The number of assignments can be different. It depends from the course program and designed by course syllabus. But the total points of the assignment are 60 in each control period.

***** Final exam can be given by instructor in the any other forms such as Written Exam, Final Quiz, Oral Tasks and Exams, etc.

Achievement level as per course curriculum shall be assessed according to the evaluation chart adopted by the academic credit system.

Letter Grade	Numerical equivalent	Percentage	Grade according to the traditional system
A	4,0	95-100	Excellent
A-	3,67	90-94	
B+	3,33	85-89	
B	3,0	80-84	Good
B-	2,67	75-79	

C+	2,33	70-74	Satisfactory
C	2,0	65-69	
C-	1,67	60-64	
D+	1,33	55-59	
D	1,0	50-54	
FX	0	25-49	Fail
F	0	0-24	

Based on the specific grade for each assignment, and the final grade, following criteria must be satisfied:

Grade	Criteria to be satisfied
90-100	- Work would be worthy of further dissemination under appropriate conditions - Mastery of advanced methods and techniques at a level beyond that explicitly taught - Ability to synthesize and employ in an original way idea from across the subject - Outstanding command of critical analysis and judgment
80-89	- Excellent range and depth of attainment of intended outcomes - Mastery of a wide range of methods and techniques - Evidence of study and originality of what has been taught - Able to display a command of critical analysis and judgement
70-79	- Attained all the intended learning outcomes for a unit - Able to use well a range of methods and techniques to come to conclusions - Able to employ critical analysis and judgement
60-69	- Some limitations in attainment of learning objectives, but has managed to grasp most of them - Able to use most of the methods and techniques taught - Evidence of study and comprehension of what has been taught but grasp insecure - Some grasp of the issues and concepts underlying the techniques and material taught, but weak and incomplete
50-59	- Attainment of only a minority of the learning outcomes - Able to demonstrate a clear but limited use of some of the basic methods and techniques taught - Weak and incomplete grasp of what has been taught - Deficient understanding of the issues and concepts underlying the techniques and material taught
25-49	- Attainment of nearly all the intended learning outcomes deficient - Lack of ability to use at all or the right methods and techniques taught - Inadequately and incoherently presented - Wholly deficient grasp of what has been taught - Lack of understanding of the issues and concepts underlying the techniques and material taught
0-24	No significant assessable material, absent or assessment missing a must pass component

5. Methodological Guidelines

Assessment is administered continuously throughout the course. The students are rated against their performance in continuous rating administered throughout the semester (60%) and summative rating done during the examination session (40%), total 100%. Continuous

rating is students' on-going performance in class and independent work. Class work is assessed for attendance, laboratory works' defense and in- class assessments.

- **ISIS (Instructor Supervised Student Independent Study)** -comprises presentation to be done by students independently and checked by instructor.
- **Mid-term and End-term** is a review of the topics covered and assessment of each student's knowledge. The form of the midterm and end term exams is complex.
- **Final assessment** is a combination of both individual (team) project (report) and oral presentation (slides) to evaluate the students' academic performance and professional skills. At the completion of this course each student has to submit an online project version conforming to the project outline, as well as to prepare and present a slide presentation that follows the presentation outline. Project iterations would be required to submit as well. Students should submit the written Report of Final project and slide-Presentation 3 days before the Final exam day.

6. Lecturer (lecturers) approvals Full name Job title Date Sign

Full name	Job title	Date	Sign
Rakhymzhan Kazbek	Assistant Professor	16.../.../2024	

Director



Sergaziyev M.Zh.